

Health Risk Classification for Insurance Premium Optimization

1. Executive Summary

Health insurance providers rely on accurate risk assessment to determine applicant eligibility and premium pricing. This project develops a machine learning-based classification system capable of predicting whether an insurance applicant is Healthy or Unhealthy using demographic, lifestyle, and medical attributes.

A complete machine learning pipeline was implemented, including data auditing, exploratory data analysis, preprocessing, model development, performance evaluation, and model selection. Three classification algorithms—Logistic Regression, Decision Tree, and Random Forest—were evaluated. Among them, Random Forest achieved the best performance with an accuracy of 85.6% and an ROC-AUC score of 0.938.

The resulting model demonstrates the potential of machine learning to support data-driven underwriting and insurance risk assessment.

2. Problem Statement

Anova Insurance seeks to optimize insurance premium pricing by assessing the health status of applicants. Health status directly influences insurance eligibility and premium determination.

The objective of this project is to build a predictive classification model that categorizes applicants as:

- Healthy (Target = 0)
- Unhealthy (Target = 1)

This classification can assist insurance providers in making informed underwriting and pricing decisions.

3. Dataset Description

The dataset contains 10,000 applicant records and 23 features describing demographic, lifestyle, and medical characteristics.

Numerical Features

- Age

- BMI
- Blood Pressure
- Cholesterol
- Glucose Level
- Heart Rate
- Sleep Hours
- Exercise Hours
- Water Intake
- Stress Level

Lifestyle and Medical Features

- Smoking
- Alcohol
- Diet
- Mental Health
- Physical Activity
- Medical History
- Allergies

Encoded Features

- Diet Type
- Blood Group

Target Variable

- 0 → Healthy
- 1 → Unhealthy

4. Data Audit

A comprehensive audit was performed to identify data quality issues.

Key Findings

- Dataset contained 10,000 observations and 23 features.
- No duplicate records were identified.
- Five columns contained missing values:
 - Blood Pressure
 - Cholesterol
 - Glucose Level
 - Medical History

- Allergies
- 96 records contained invalid Age values equal to zero.
- Target classes were almost perfectly balanced:
 - Healthy: 50.01%
 - Unhealthy: 49.99%

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RangeIndex: 10000 entries, 0 to 9999
Data columns (total 23 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                    10000 non-null  float64
1   BMI                                    10000 non-null  float64
2   Blood_Pressure                        9471 non-null   float64
3   Cholesterol                           9462 non-null   float64
4   Glucose_Level                         9486 non-null   float64
5   Heart_Rate                            10000 non-null  float64
6   Sleep_Hours                           10000 non-null  float64
7   Exercise_Hours                        10000 non-null  float64
8   Water_Intake                           10000 non-null  float64
9   Stress_Level                           10000 non-null  float64
10  Target                                10000 non-null  int64
11  Smoking                               10000 non-null  int64
12  Alcohol                               10000 non-null  int64
13  Diet                                   10000 non-null  int64
14  MentalHealth                           10000 non-null  int64
15  PhysicalActivity                       10000 non-null  int64
16  MedicalHistory                         9465 non-null   float64
17  Allergies                             9479 non-null   float64
18  Diet_Type_Vegan                        10000 non-null  bool
19  Diet_Type_Vegetarian                   10000 non-null  bool
20  Blood_Group_AB                         10000 non-null  bool
21  Blood_Group_B                          10000 non-null  bool
22  Blood_Group_O                          10000 non-null  bool
dtypes: bool(5), float64(12), int64(6)
memory usage: 1.4 MB

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Figure 1: Dataset Overview and Missing Value Analysis

5. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand feature distributions and relationships with the target variable.

Univariate Analysis

Histograms and boxplots were generated for all numerical variables.

Key observations:

- BMI exhibited a near-normal distribution.
- Blood Pressure and Cholesterol contained several outliers.
- Age included 96 records with invalid values equal to zero.
- Most categorical variables displayed balanced distributions.

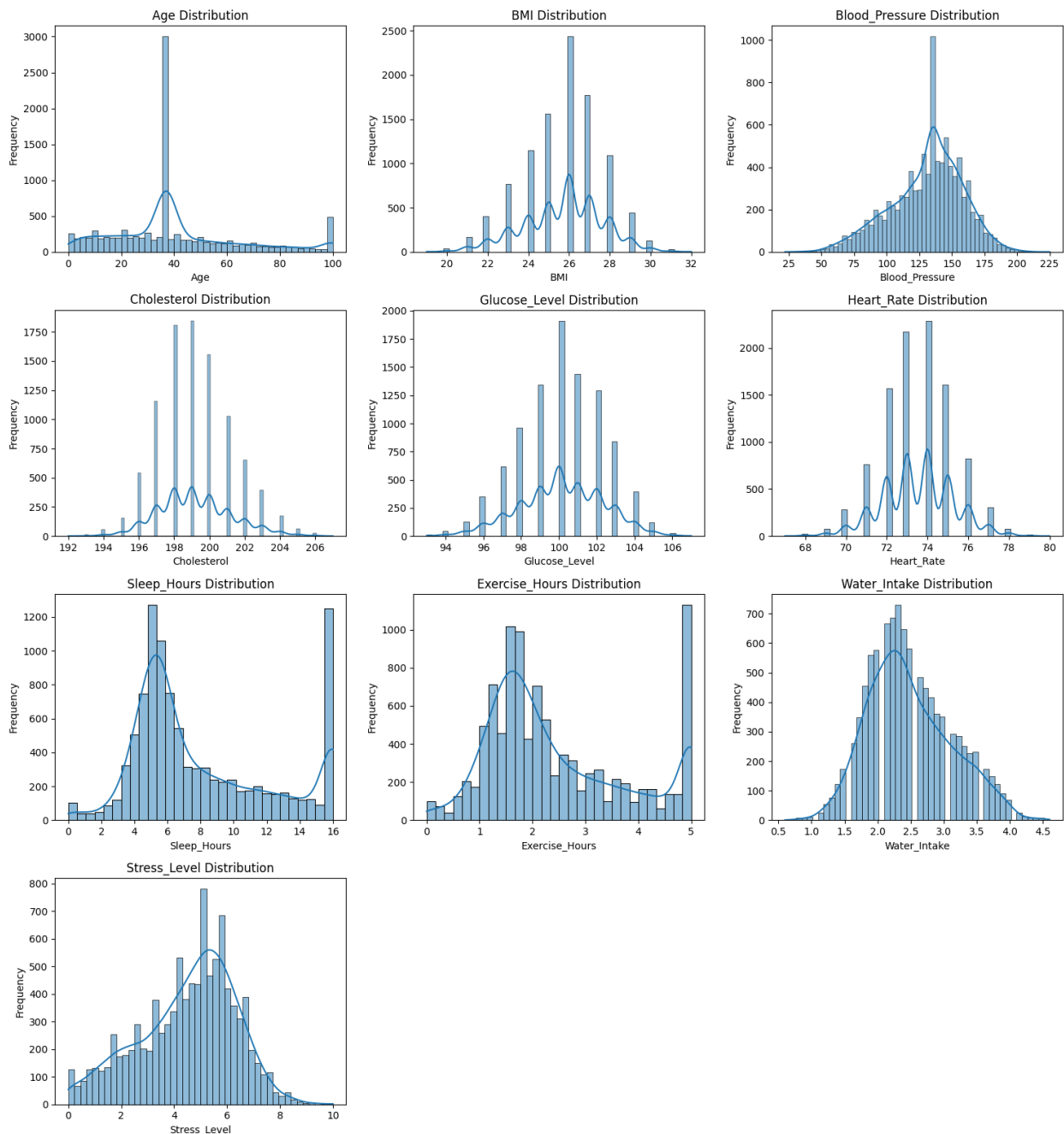


Figure 2: Numerical Feature Distributions

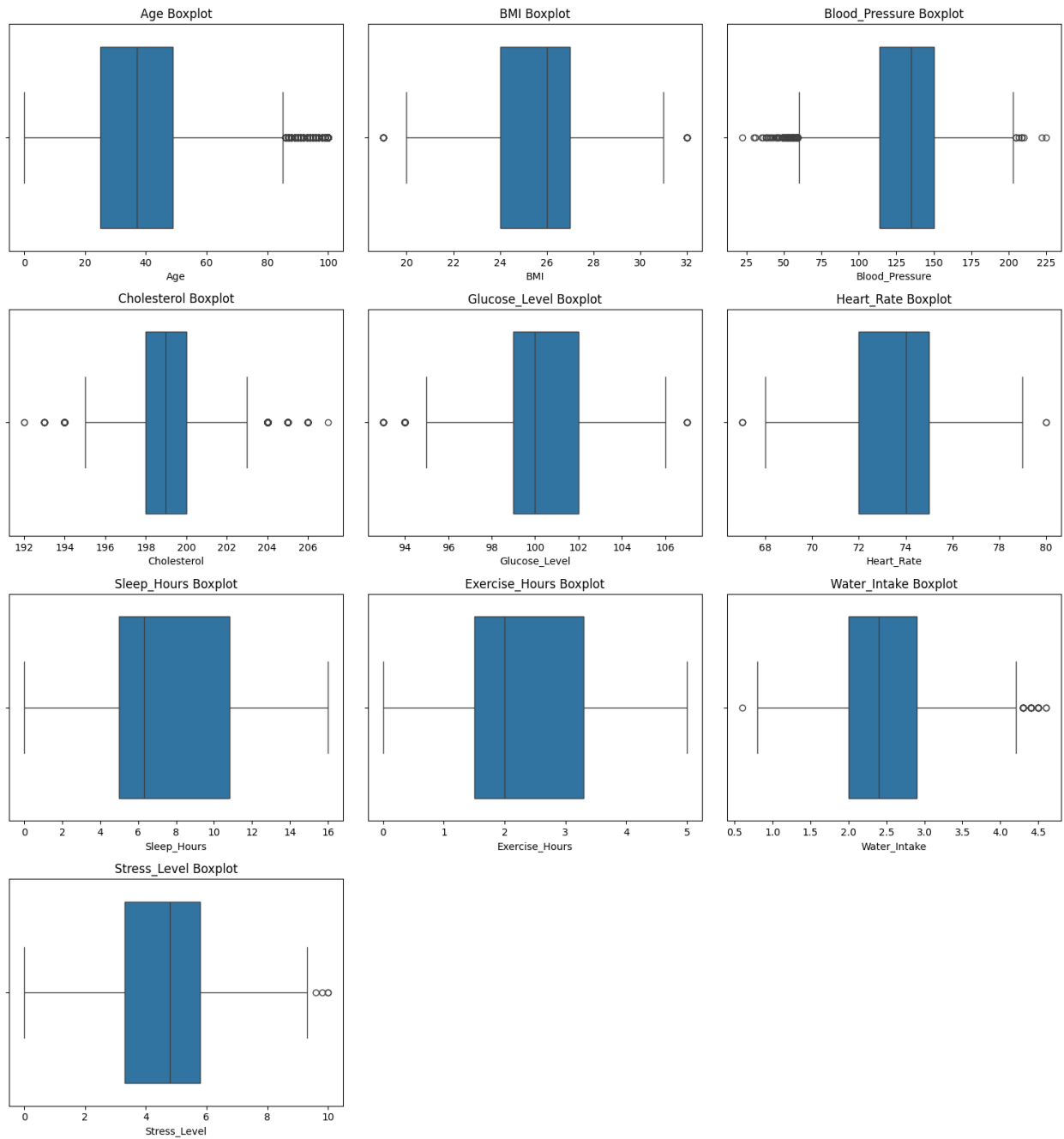


Figure 3: Numerical Feature Boxplots

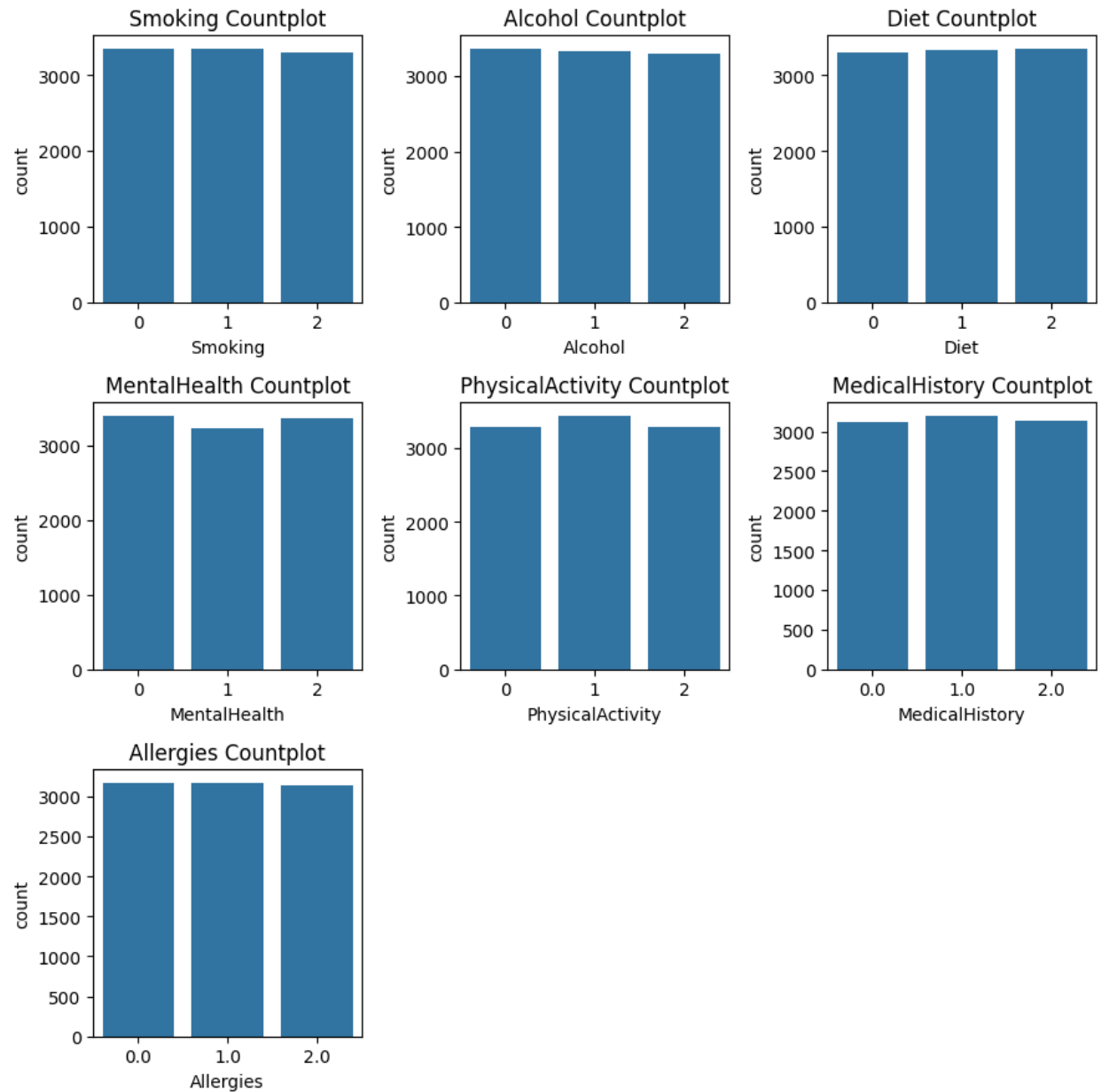


Figure 4: Categorical Feature Count Plots

6. Bivariate Analysis

Relationships between individual features and the target variable were investigated.

Key Findings

- BMI demonstrated the strongest positive association with health risk.
- Stress Level showed a moderate positive relationship with the target.

- Blood Pressure, Cholesterol, and Age exhibited significant associations with health status.
- Lifestyle variables such as Smoking and Alcohol showed limited standalone predictive power.

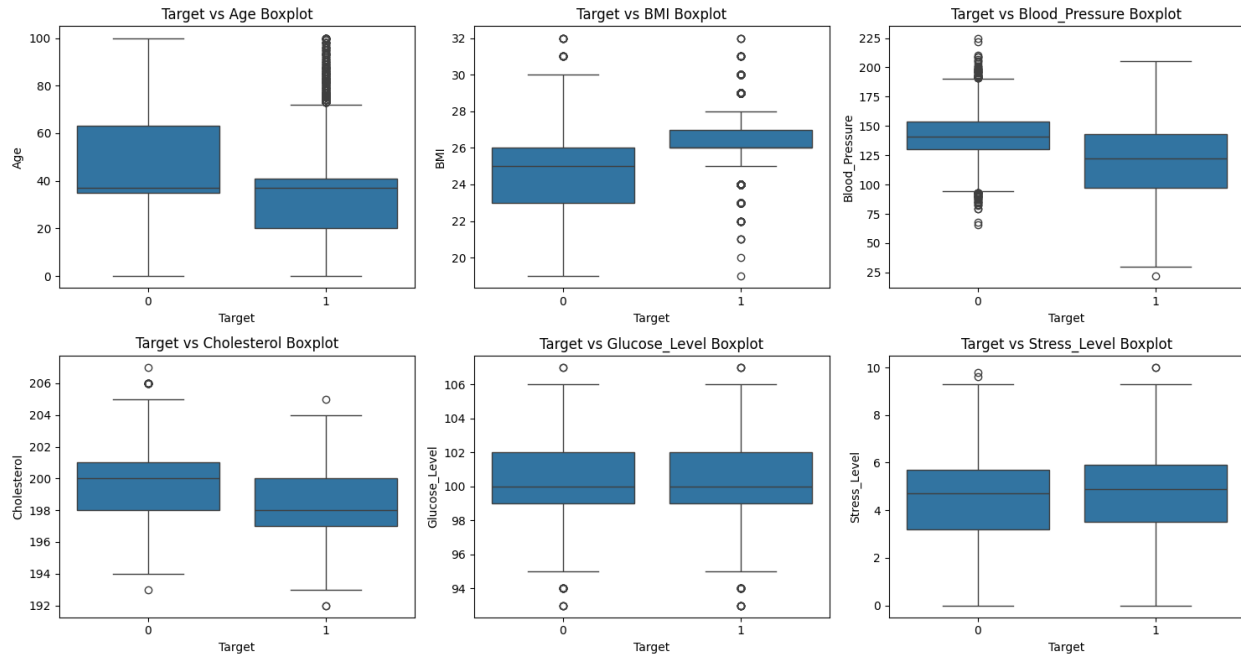


Figure 5: Target vs Numerical Features Boxplots

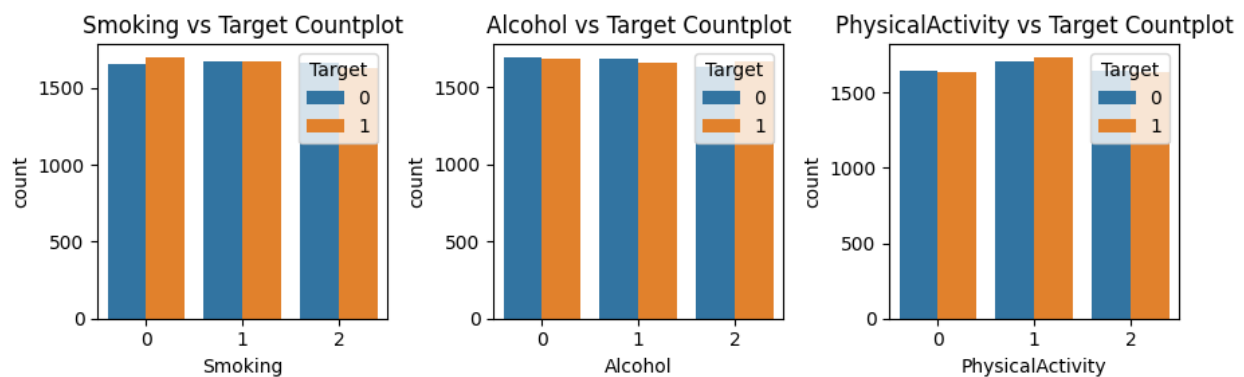


Figure 6: Target vs Categorical Features Count Plots

7. Data Preprocessing

Several preprocessing steps were performed to prepare the data for modeling.

Invalid Age Handling

- 96 records containing Age = 0 were treated as invalid observations.
- These records represented less than 1% of the dataset and were removed.

Missing Value Treatment

Median Imputation:

- Blood Pressure
- Cholesterol
- Glucose Level

Mode Imputation:

- Medical History
- Allergies

Final Dataset

- Original Records: 10,000
- Final Records: 9,904
- Missing Values Remaining: 0

8. Model Development

Three machine learning algorithms were developed and evaluated.

Logistic Regression

A Logistic Regression model was trained as the baseline classifier.

Results:

- Accuracy: 78.0%
- ROC-AUC: 0.862

Decision Tree

A Decision Tree classifier was trained to capture non-linear relationships.

Results:

- Accuracy: 80.6%
- ROC-AUC: 0.806

Training accuracy reached 100%, indicating overfitting.

Random Forest

A Random Forest classifier was trained as an ensemble approach to improve generalization.

Results:

- Accuracy: 85.6%
- ROC-AUC: 0.938

The Random Forest model demonstrated the best overall performance.

9. Model Evaluation and Comparison

Performance Comparison

Model	Accuracy	ROC-AUC
Logistic Regression	0.780	0.862
Decision Tree	0.806	0.806
Random Forest	0.856	0.938

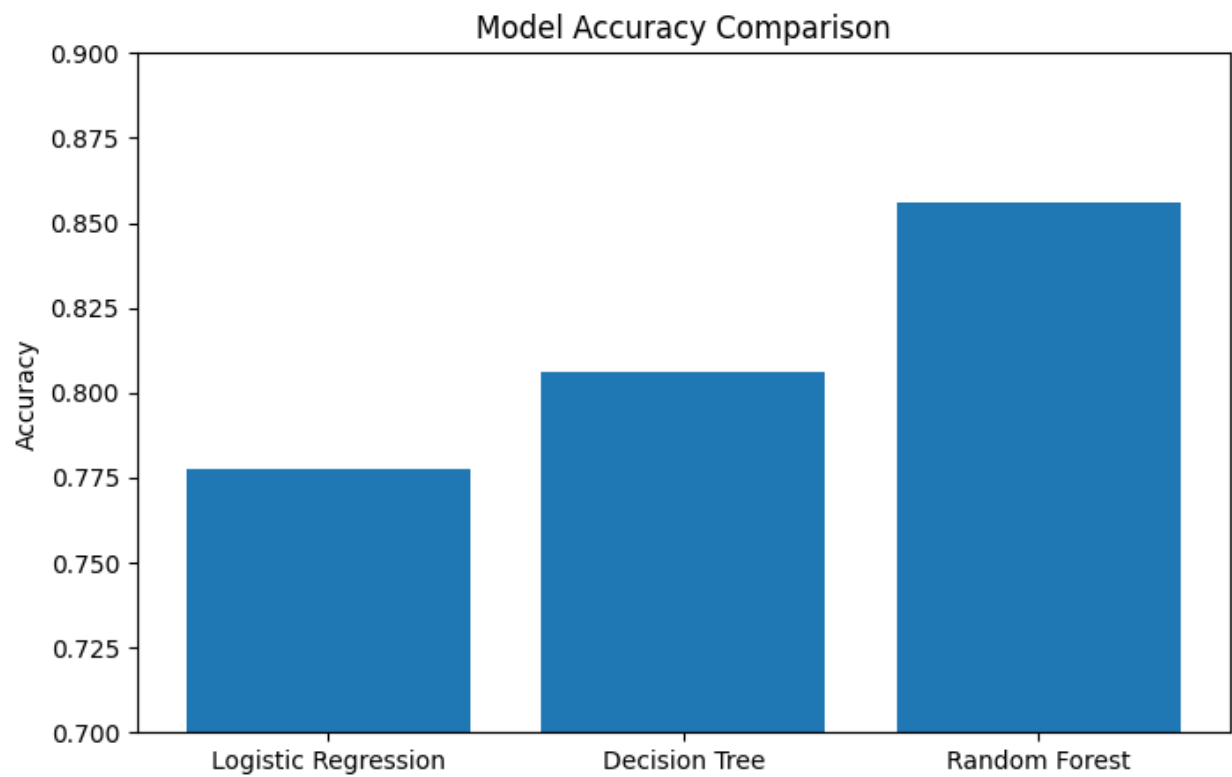


Figure 7:Model Accuracy Comparison

ROC-AUC Analysis

ROC curves were generated for all three models.

Key findings:

- Random Forest achieved the highest ROC-AUC score.
- Logistic Regression showed strong discrimination capability despite lower accuracy.
- Decision Tree exhibited the weakest ROC performance.

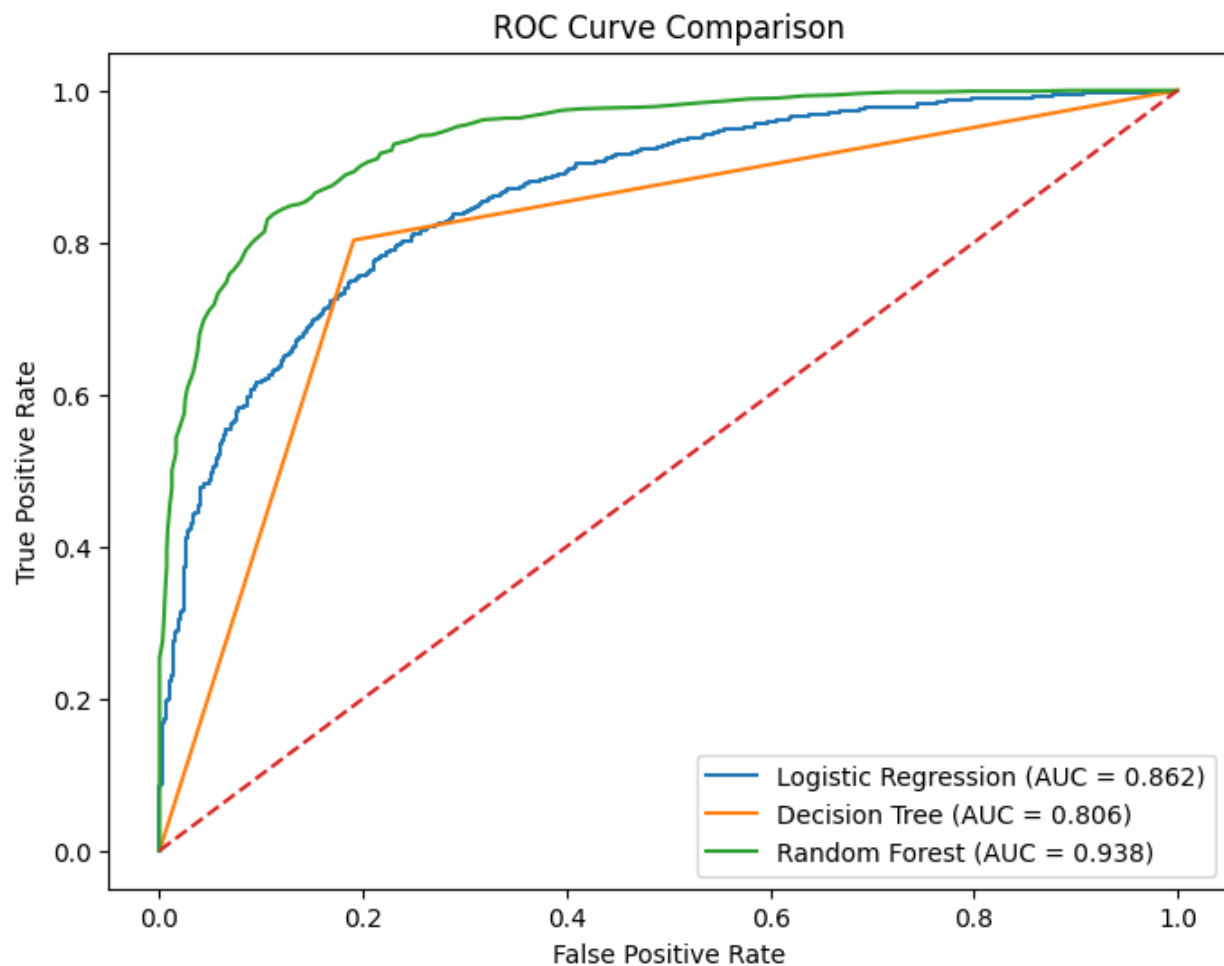


Figure 8: ROC Curve Comparison

10. Learning Curve Analysis

Learning curve analysis was performed for the Random Forest model.

Observations

- Training accuracy remained close to 100%.
- Validation accuracy steadily improved with increasing training data.
- The model demonstrated good generalization capability.
- Additional training data may further improve performance.

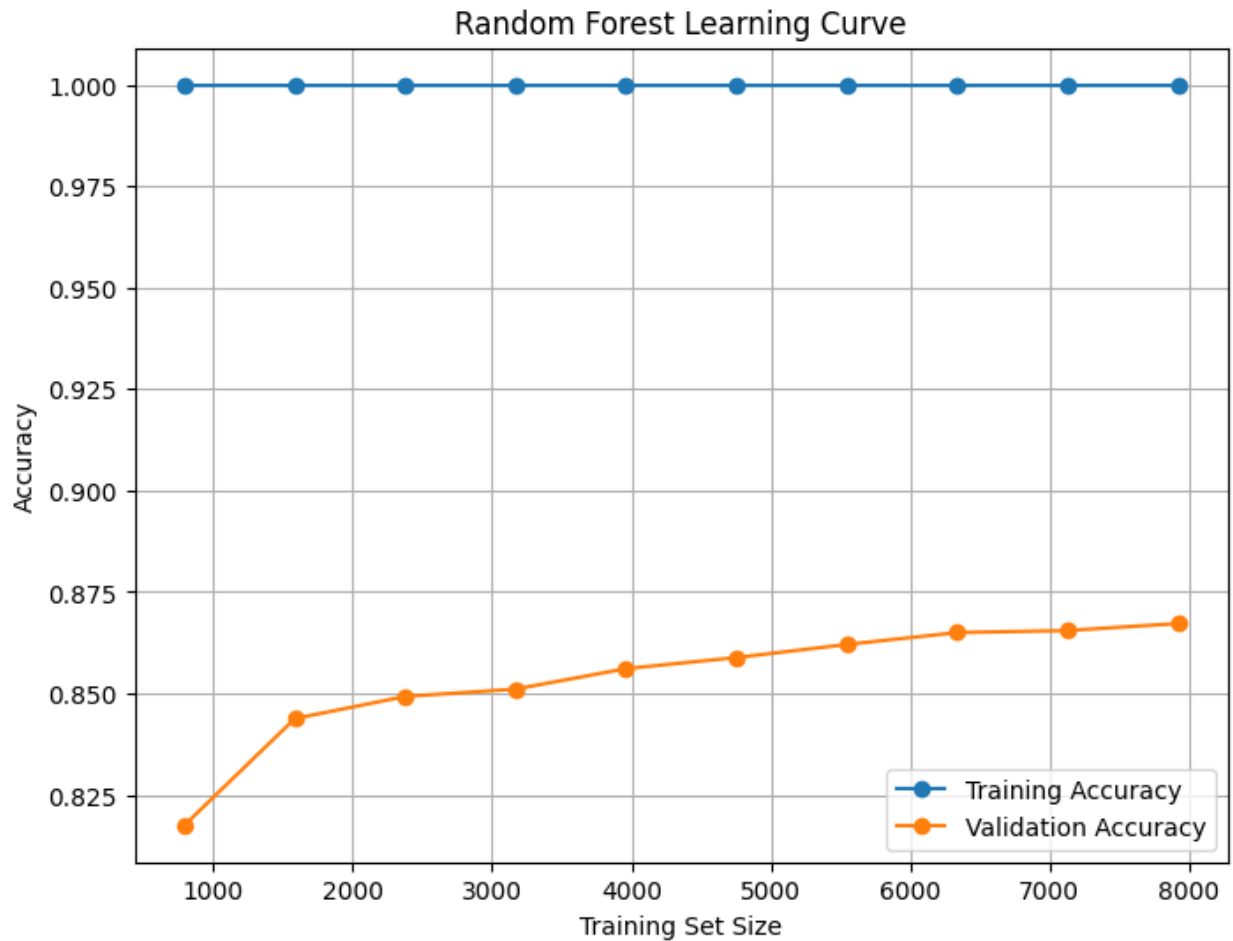


Figure 9: Random Forest Learning Curve

11. Feature Importance Analysis

Feature importance scores from the Random Forest model identified the most influential predictors.

Top Predictive Features

1. BMI
2. Blood Pressure
3. Cholesterol

4. Age
5. Glucose Level

These variables contributed most significantly to health-risk classification.

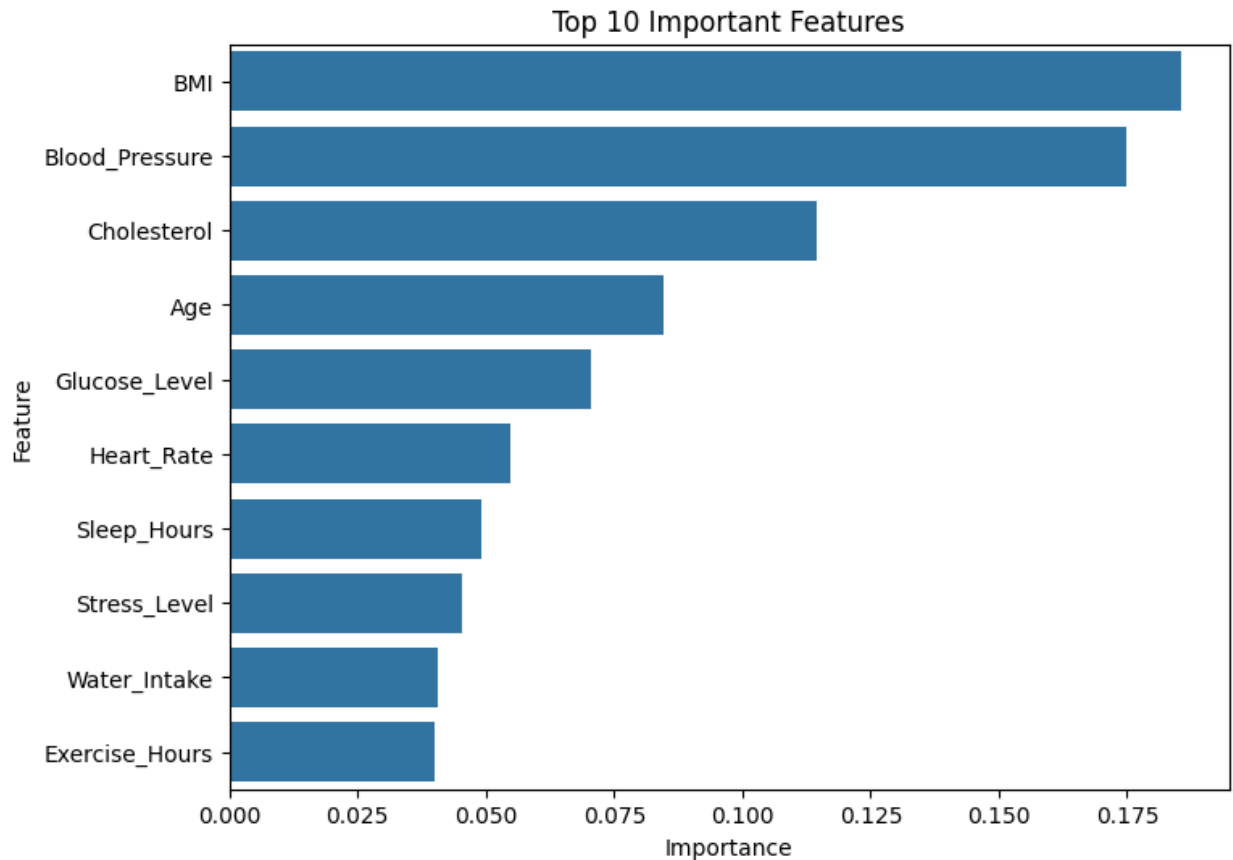


Figure 10: Random Forest Feature Importance

12. Business Recommendations

Based on the model findings:

Risk-Based Premium Pricing

Applicants exhibiting unfavorable values in BMI, Blood Pressure, Cholesterol, and Glucose Level may be considered for higher premium plans or additional medical evaluation.

Early Risk Detection

The developed model can assist underwriters in identifying potentially high-risk applicants during the application process.

Personalized Health Programs

Insurance providers may introduce wellness programs targeting modifiable risk factors such as obesity, physical inactivity, and stress.

Data-Driven Underwriting

Machine learning-based risk assessment can improve consistency, scalability, and objectivity in underwriting decisions.

13. Conclusion

This project successfully developed a machine learning solution for insurance health-risk classification.

A complete workflow was implemented, including:

- Data auditing
- Exploratory data analysis
- Data preprocessing
- Model development
- Performance evaluation
- Model comparison
- Feature importance analysis

Among the evaluated models, Random Forest delivered the strongest performance with an accuracy of 85.6% and an ROC-AUC score of 0.938.

The results demonstrate that machine learning can effectively support insurance premium pricing and risk assessment by leveraging health-related applicant data.

14. Future Scope

Potential future improvements include:

- Hyperparameter tuning of ensemble models
- Integration of additional medical indicators
- Deployment as a real-time underwriting support tool
- Explainable AI implementation using SHAP values
- Continuous model retraining with updated applicant data